



ALLEN
CAREER INSTITUTE
KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
07-03-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

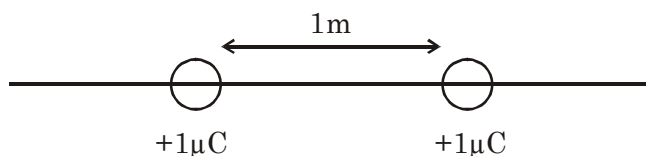
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. On a smooth horizontal wire, we have 2 small charged beads of charge $1\mu\text{C}$ and mass 1gm each. The bead on right is released and when the distance between them increases to 2m , the left bead is also released. In the motion after that,



- (A) The velocity of center of mass is 1.5 m/s to right.
- (B) The velocity of right bead after a long time is $\frac{3(\sqrt{3}+1)}{2} \text{ m/s}$ to right.
- (C) The velocity of left bead after a long time is $\frac{3(\sqrt{2}+1)}{2} \text{ m/s}$ to left.
- (D) When the speed of left bead is 1 m/s , the distance between them is 4 m .

Space for Rough Work

2. A uniformly charged thin spherical shell of radius a and charge $+Q$ is enclosed by another uniformly charged thin spherical shell of radius $2a$ and charge $+2Q$. Both shells are non conducting in nature.

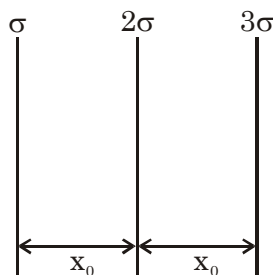
(A) The total potential energy of system is independent of position of small shell inside the larger shell.

(B) If the centre of inner shell is at a distance $\frac{a}{2}$ from centre of outer shell, the force on outer shell is $\frac{8kQ^2}{a^2}$.

(C) Self energy of inner shell is half that of the outer shell.

(D) The pressure on inner shell is $\frac{Q^2}{32\pi^2 a^4 \epsilon_0}$.

3. Three large sheets of mass/area ρ and charge per unit area as shown are located in gravity free space. When they are released,



(A) The middle sheet accelerates to left with an acceleration of $\frac{4\sigma^2}{\rho \epsilon_0}$.

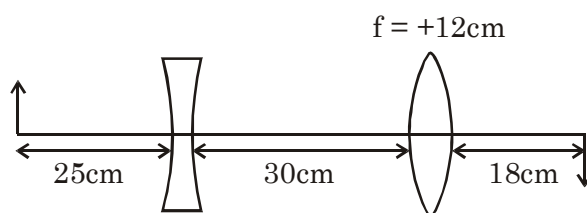
(B) The left sheet accelerates to left with an acceleration of $\frac{5\sigma^2}{2\rho \epsilon_0}$.

(C) The right plate accelerates to right with an acceleration of $\frac{7\sigma^2}{\rho \epsilon_0}$.

(D) The middle plate does not accelerate because centre of mass remains at rest.

Space for Rough Work

4. A metallic body has a cavity in it. A charge is suspended inside the cavity. There is no net charge on the metallic body.
- (A) If the outer surface of metallic body is of shape of a sphere, the electric field outside the sphere can be found even if cavity is of irregular shape.
- (B) If the outer surface of metallic body is shaped like a sphere, electric field outside the sphere can be found only if cavity is also spherical in shape.
- (C) Even if the outer surface is irregular, electric field outside the metallic body can be found if the cavity is spherical in shape.
- (D) The electric field outside the metallic body can be found even if the cavity & outside surface are both irregular in shape.
5. A real inverted image is formed of a real object due to an optical system shown (in air).



- (A) The focal length of diverging lens is $-\frac{150}{19}\text{cm}$
- (B) The magnification due to the system is -0.12
- (C) If the diverging lens is moved slightly to the right, image moves to the right.
- (D) If the converging lens is moved slightly to the left, image moves to the left.

Space for Rough Work

6. A particular star has two planets in a coplanar orbits. The radius of the orbit of the first planet is R and that of the other is $4R$. The time period of revolution of the first planet around the star is one year.
- (A) The time period of the second planet is 8 years.
- (B) If both the planets are revolving in the same sense and at $t = 0$, planets and star are all along the same line, they will be in the same line at $t = 8/7$ years
- (C) If both the planets are revolving in the opposite sense and at $t = 0$, planets and star are all along the same line, they will be in the same line at $t = 8/7$ years
- (D) If both the planets are revolving in the opposite sense and at $t = 0$, planets and star are all along the same line, the planets will be at a distance of $5R$ at $t = 4/9$ years

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Before the advent of solid-state electronics, vacuum tubes were widely used in radio and other devices. A simple type of vacuum tube known as a diode consists essentially of two electrodes within a highly evacuated enclosure. One electrode, the cathode, is maintained as a high temperature and emits electrons from its surface. A potential difference of a few hundred volts is maintained between the cathode and the other electrode, known as the anode, with the anode at the higher potential.

A vacuum tube diode consists of concentric cylindrical electrodes, the negative cathode and the positive anode. Because of the accumulation of charge near the cathode, the electric potential, between the electrodes is not a linear function of the position, even with planar geometry, but is given by

$$V(x) = Cx^{4/3}$$

where x is the distance from the cathode and C is a constant, characteristics of a particular diode and operating conditions. Assume that the distance between the cathode and anode is 27.0 mm and the potential difference between electrodes is 243 V.

7. Determine the value of C .
 (A) $6 \times 10^4 \text{ V/m}^{4/3}$ (B) $3 \times 10^4 \text{ V/m}^{4/3}$ (C) $2 \times 10^4 \text{ V/m}^{4/3}$ (D) None of these
8. Determine the force on an electron when the electron is 8mm from cathode.
 (A) $1.28 \times 10^{-15} \text{ N}$ (B) $3.24 \times 10^{-15} \text{ N}$ (C) $6.28 \times 10^{-15} \text{ N}$ (D) $8.24 \times 10^{-15} \text{ N}$

Space for Rough Work

Paragraph for Questions 9 and 10

Recent studies of the planet pluto of the solar system shows the following data: (1) Pluto is observed from Earth in the form of a small disk with angular diameter 0.24 arcsec, (2) Pluto has a plutoosynchronous satellite at a height of 860 km from its surface (stationary as seen from the surface of pluto), (3) Pluto's orbital period around its axis is equal to 10^6 seconds and (4) the distance from Earth to Pluto is approximately equal to 5.4×10^9 km. On the basis of these data

9. Find The average density of the planet Pluto is closest to
(A) 1 kg/m^3 (B) 100 kg/m^3 (C) 10 kg/m^3 (D) 1000 kg/m^3
10. Find the approx radius of orbit of pluto's satellite described here.
(A) 4000 km (B) 2000 km (C) 6400 km (D) 9600 km

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

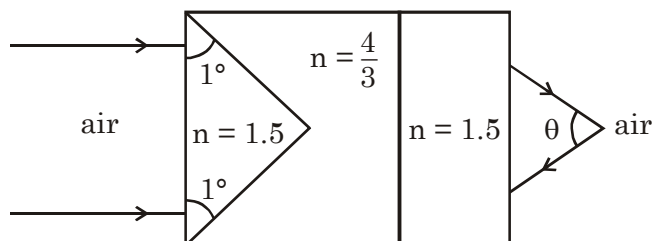
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										-	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A fountain designed to spray a column of water, 12 m into the air has a 1 cm diameter nozzle at ground level. The water pump is 3m below the ground. The pipe to the fountain has a diameter of 2 cm. Find the necessary gauge pressure at the pump (in 10^5 Pa). Neglect the viscosity of water.
2. Two parallel rays are incident on a biprism (combination of two prisms) in what is the angle θ (in degrees) between the two emergent rays in air ?



Space for Rough Work

3. A fish hides under a floating plank which is circular in shape with radius $R = 3$ m. The index of refraction of water is $\sqrt{2}$. Find the volume of water (in m^3) in which it can swim without being detected from above.



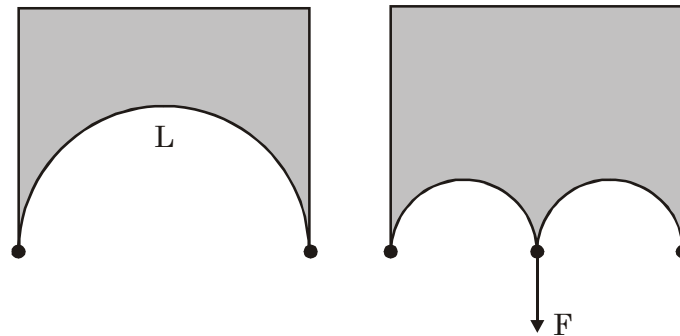
4. You are given a long, thin polished metal tube and asked to put your eye at one end and to point the other end at a diffuse light source (a well-illuminated wall, for example). What you see is a pattern of rings. At the centre there is a bright circle, which is the diffuse light source seen directly through the hole in the end of the tube. Surrounding this is a ring of light that is less bright than the central circle. Around that there's another ring that's less bright still. And so on. The rings become successively less bright until the pipe appears dark. The rings appear to have approximately the same width as they get larger. You are asked to derive an equation for the angle subtended at the eye by the n -th ring. If the diameter of the tube is 2cm and length of the tube is 1 m, what is the angle subtended at the eye by the 5th ring (in radians).

Space for Rough Work

5. A solid sphere of mass M of radius R is moving with speed $\sqrt{\frac{GM}{R}} = 100\sqrt{10} \text{ ms}^{-1}$ in free space. Due to application of external impulsive force, it breaks into eight spheres of same radius in such a way that the total mechanical energy before breaking and after breaking of sphere remains same. The speed of each sphere when separation between them is very large (assume at that instant speed of each sphere is same) is given by _____ m/s.
6. The diameter of a standard table-tennis ball is 40 mm, and its mass is 3.14 grams. What is the radius (in cm) of that iron ball (density 8000 kg/m^3), which should be attached to the table-tennis ball, in order that they just float in the water (density 1000 kg/m^3)?

Space for Rough Work

7. In a soap film experiment an originally semicircle shaped thread of length L can be deformed to two semicircles with a force of F exerted at the middle of the thread. Calculate the surface tension. $F = 1\text{N}$, $L = 20\text{ cm}$.



8. The velocity distribution over a flat plate is given by $u = \frac{3}{4}y - y^2$, where u is the velocity in m/s at a distance y (in m) above the plate and μ for the given fluid is 0.84 Ns/m^2 . The shear stress at location 0.3 m above the plate is k times the shear stress at a location 0.2 m above the plate. The value of k would be

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 24)

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
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1. The reaction : $2A(g) \rightarrow 3B(g) + C(l)$; follows first order kinetics. The following data is obtained at constant temperature, starting with pure A(g) only

Time (min.)	0	10	∞
Total pressure (bar)	0.2	0.3	0.38

The correct information(s) about the reaction and reaction components is/are -

(A) The vapour pressure of C(l) is 0.08 bar.

(B) 50% reaction occur in 10 min.

(C) Rate constant for the reaction is $\left(\frac{\ln 2}{20}\right) \text{ min}^{-1}$

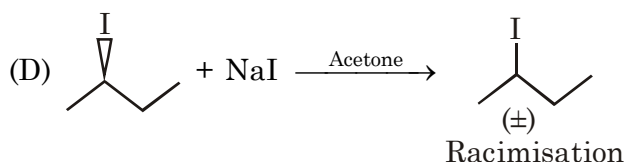
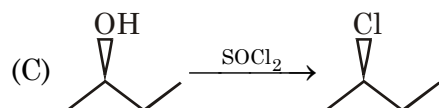
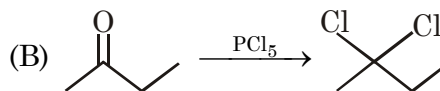
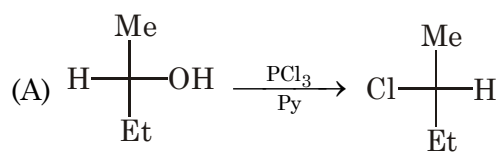
(D) At $t = 8$ min, the pressure exerted by the vapours of C(l) = 0.08 bar.

Space for Rough Work

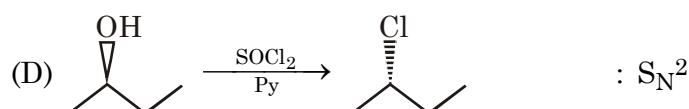
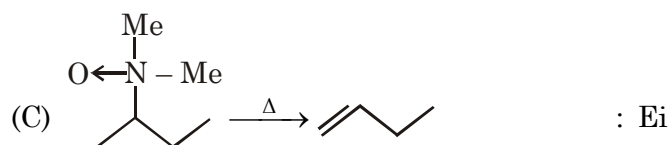
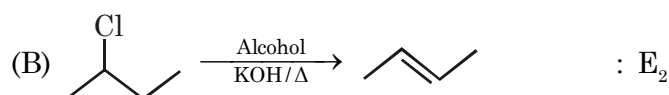
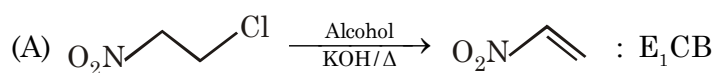
2. 0.4 mole glucose (non-volatile) is dissolved in 500 gm water to form an ideal solution. When the solution is cooled to -5.4°C , the correct information(s) about final solution is/are -
[Given : for water : Normal freezing point = $0^{\circ}\text{C} = 273\text{ K}$ and $k_f = 1.8\text{ K.kg/mol}$]
- (A) Entire liquid solution will convert into solid solution
- (B) Ice deposited over the liquid solution will have glucose and water molecules in 18 : 125 mass ratio.
- (C) $\frac{1100}{3}$ gm liquid water will separate as ice.
- (D) For final liquid solution, mole-fraction of glucose = $\frac{27}{527}$
3. Select **CORRECT** statement/s for $\text{Cu}_2^{\text{II}}[\text{Fe}(\text{CN})_6]$
- (A) It is coloured compound.
- (B) It is paramagnetic species.
- (C) Axial d-orbitals are used in hybridization of central metal ion.
- (D) Non-axial d-orbitals are used in hybridization of central metal ion.
4. Which of the following is/are **CORRECT** statement(s)?
- (A) Ligands with vacant π^* molecular orbitals are also known as π -acid ligands.
- (B) π -acceptor ligands can accept electron cloud from central metal atom / ion.
- (C) Delocalization of electrons from metal to ligand increase M-C bond order.
- (D) Synergic bonding increases the M-C bond length.

Space for Rough Work

5. Identify reactions correctly matched with their major product?



6. Identify reactions correctly matched with their mechanism?



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

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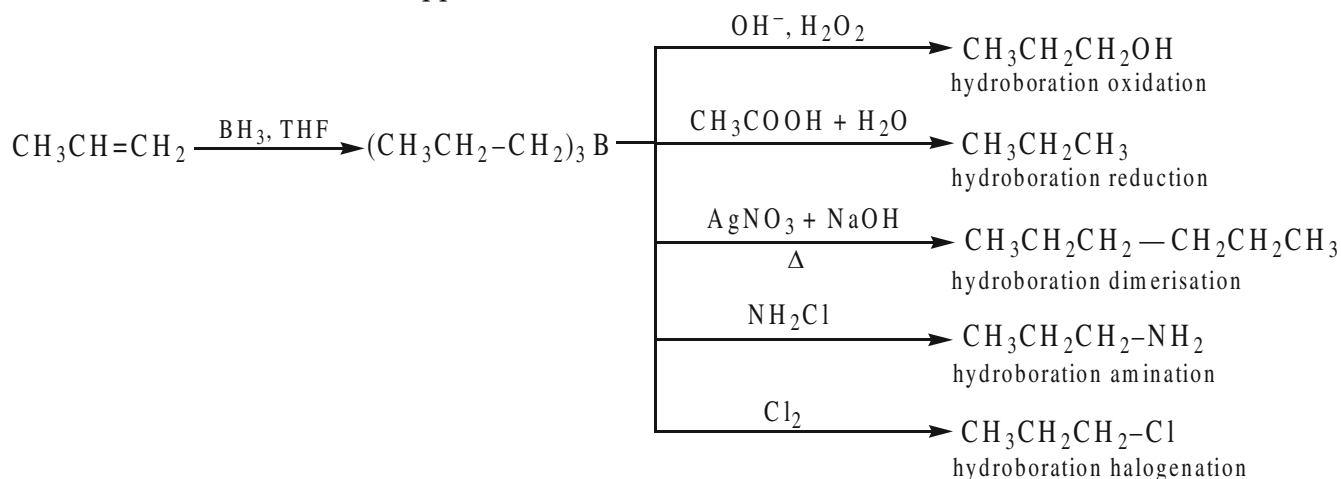
Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

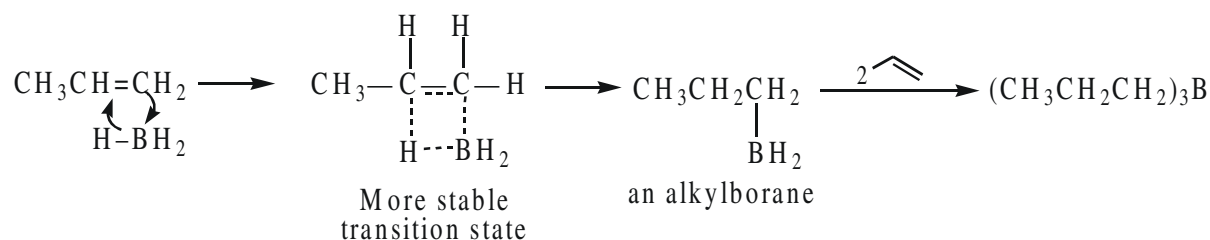
Paragraph for Questions 7 and 8

HYDROBORATION-OXIDATION

Hydroboration has been developed by brown as a reaction of tremendous synthetic utility because alkyl boranes are able to undergo a variety of transformation. Hydroboration is a one step, four centre, cis addition process in accordance with Markovnikov's rule but after oxidation it seems to be appear to violate Markovnikov's rule.

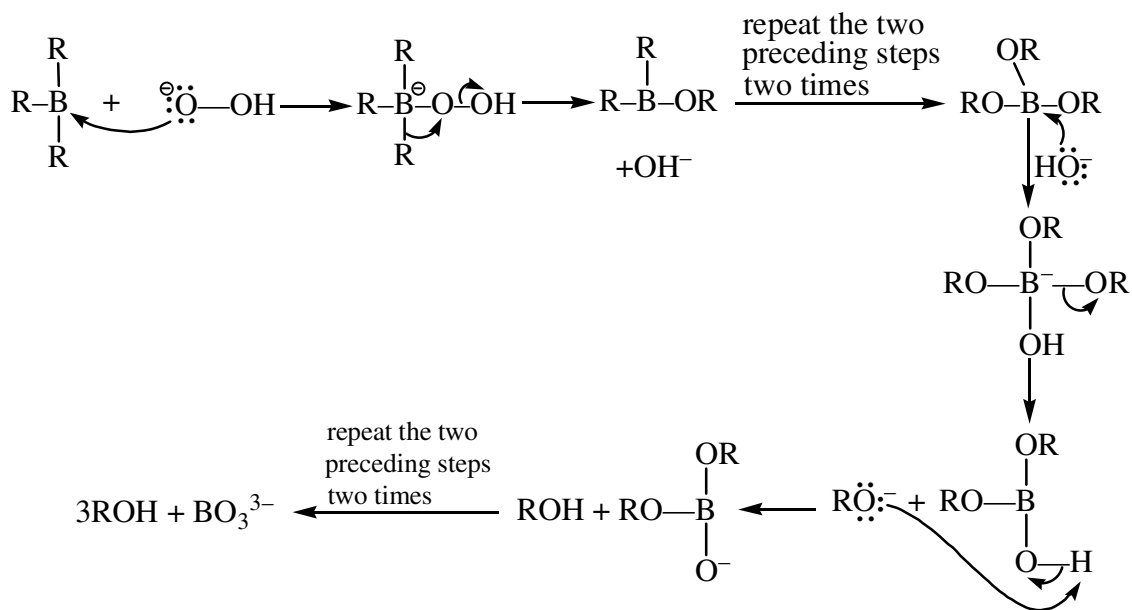
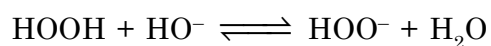


Mechanism of Hydroboration :

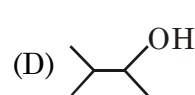
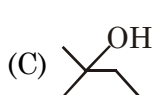
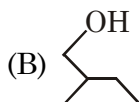
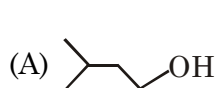
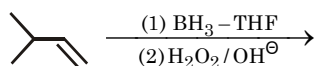


Space for Rough Work

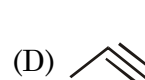
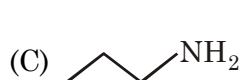
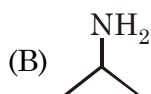
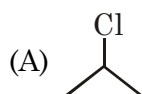
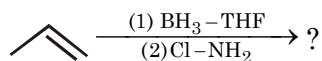
Mechanism of oxidation :



7. Identify major product of following reaction



8. Identify major product of following reaction?



Space for Rough Work

Paragraph for Questions 9 and 10

The nature of complex compounds was first studied by Werner. He was able to explain the nature of bonding in complexes and he concluded that in complexes, the metal shows two different type of valencies, viz. Primary & secondary valencies.

Separately, Pauling developed the Valence bond theory which considered that different atomic orbitals on the metal are used for bonding. Metal must have empty orbital of suitable energy & ligand must have a lone pair of electrons.

9. Which of the following octahedral complexes show maximum electrical conductivity in aqueous solution.

(A) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (B) $\text{CoCl}_3 \cdot 6\text{NH}_3$ (C) $\text{PtCl}_4 \cdot 2\text{NH}_3$ (D) $\text{CoCl}_3 \cdot 3\text{NH}_3$

10. Which of the following outer-orbital complexes are paramagnetic.

(A) $[\text{Zn}(\text{NH}_3)_4]\text{SO}_4$ (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (C) $\text{Na}_3[\text{FeF}_6]$ (D) $[\text{Cr}(\text{en})_3]\text{Cl}_3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A solid metal oxide (M_2O_3) crystallises as HCP for O^{2-} ions and the metal ions (M^{3+}) occupy only octahedral voids in ideal crystal. The metal may exhibit +3 and +4 oxidation states. The composition of a real metal oxide crystal is found to be $M_{0.60}O_{1.00}$. The percentage of metal atoms missing in the crystal is x% and the ratio of numbers of M^{3+} and M^{4+} ions in the crystal is y : 1, then the value of (y/x) is.
2. Liquids 'A' and 'B' form an ideal solution. A solution is made by mixing 4 moles of liquids 'A' and 6 moles of liquid 'B' and pressure over the system is maintained to 55 cm Hg. The mole-fraction of 'A' in liquid solution at 55 cm Hg is.

(Given : $P_A^0 = 40 \text{ cmHg}$, $P_B^0 = 60 \text{ cmHg}$)

Space for Rough Work

3. 60% of $C_2H_2(g)$ molecules trimerise to $C_6H_6(g)$ molecules at 400 K and 0.3 atm. The value of $k_p(\text{atm}^{-2})$ of reaction : $3C_2H_2(g) \rightleftharpoons C_6H_6(g)$, is
4. 20% of first order reaction complete in 20 min. The percentage completion of reaction in 60 min. is (Temperature remains constant)
5. As per C.F.T. Cl–Mn–Cl angle in $[MnCl_4]^{2-}$ is $109^\circ 28'$ what is the angle between e-orbital of central metal and the ligand is –
(If your answer is $84^\circ 32'$ then you have to fill 84.32 in OMR)

Space for Rough Work

6. From the following list of ligands, find the value of $\frac{(X+Y)}{Z}$

gly⁻, NO₂⁻, S₂O₃²⁻, , NO₃⁻, SCN⁻, acac⁻, dipy

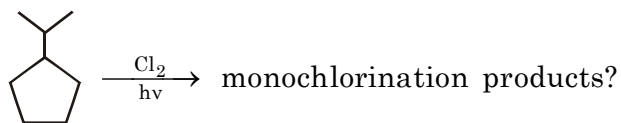
where, X = Total number of bidentate ligands that form five membered ring with central metal atom ion.

Y = Total number of π -acid ligands that can form π -complex

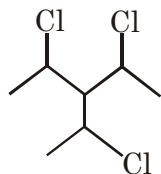
Z = Total number of ambidentate ligand(s) with flexidentate character

Space for Rough Work

7. Identify total number of monochlorination products in following reaction?



8. Identify total number of stereoisomers for following compound?



Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to-

(A) $\frac{4}{3}$

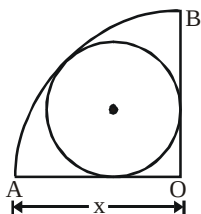
(B) $-\frac{4}{3}$

(C) $\frac{8}{3}$

(D) $-\frac{8}{3}$

Space for Rough Work

2. Sector OAB is a quarter of circle of radius x . A circle of radius ' r ' is drawn inside this sector tangent to three points as shown, then which of the following is/are true ? (where r & x are variable)



- (A) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 + 2\frac{d^2x}{dr^2} = 6$ (B) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 - 2\frac{d^2x}{dr^2} = 8$
- (C) $\frac{x+r}{x-r} + \frac{r-x}{x+r} = 2$ (D) $\sum_{n=1}^{10} \frac{d^n r}{dx^n} = \frac{r}{x}$
3. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \quad \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2^x$ is
- (A) 4 (B) 3 (C) 2 (D) 1
4. Let $f(x) = \begin{cases} -2 & \text{for } -3 \leq x \leq 0 \\ x-2 & \text{for } 0 < x \leq 3 \end{cases}$, $g(x) = [f(|x|)] + [|f(x)|]$ where $[.]$ denotes the greatest integer function. Which of the following statement(s) is/are true ?
- (A) $g(1.7) = -1$
- (B) $g(x)$ is not differentiable at any point
- (C) $g(x)$ is continuous at all non integral points
- (D) $g(x)$ is differentiable at some points

Space for Rough Work

5. Let $f(x)$ be a derivable function on $\mathbb{R}$ such that its derivative function is given by

$$f'(x) = \frac{4x^2 - x^4 - 1}{2(x^4 + 1)}, \text{ then}$$

- (A) $f(x)$ is concave up for $x \in (-\infty, -1) \cup (0, 1)$
 (B) $f(x)$ has exactly one point of local maxima
 (C) $f(x)$ has an inflection point at $x = -1, 0, 1$
 (D) $f(x)$ has exactly two points of local maxima
6. Consider $f(x) = (x-1)\tan^{-1}x - \frac{1}{2}\ln(1+x^2), x \geq 1$. Identify which of the following statement(s) is(are) correct ?

(A) $\lim_{x \rightarrow \infty} f'(x) = \frac{\pi}{2}$

(B) $f'(x)$ is increasing in $[1, \infty)$

(C) $f(x+4) - f(x) > 2\pi \forall x \in [1, \infty)$

(D) $f(x+4) - f(x) < 2\pi \forall x \in [1, \infty)$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

$$x^2 - 2(1 + 2k)x + 8 + k = 0, \text{ where } k \in \mathbb{R}.$$

7. If both roots of given equation are positive, then -

(A) $k \in \left(-\infty, -\frac{7}{4}\right] \cup [1, \infty)$

(B) $k \in (-8, \infty)$

(C) $k \in \left(-\frac{1}{2}, \infty\right)$

(D) $k \in [1, \infty)$

8. If roots are in ratio 1 : 2, then the value of $(32k^2 + 23k)$ is equal to

(A) 36

(B) 48

(C) 64

(D) 72

Space for Rough Work

Paragraph for Questions 9 and 10

Let $g : \mathbb{R} \rightarrow \left\{ \frac{1}{2} \right\}$ be a function given by $g(x) = x^{10}(f'(t) - 2) + x^9(f''(t)) + 4xf(0) + \frac{1}{2}$ where $t \in \mathbb{R}$.

9. Number of solutions of $|f(x)| = \cos x$ is -

- (A) 0 (B) 1 (C) 2 (D) infinite

10. Number of points of non differentiability of function $h(x) = |f(x)(e^{f(x)} - 1)(f(x) - 1)(f(x) - 2)|$ is

- (A) 2 (B) 3 (C) 4 (D) 5

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If equation $x^2 + ax + b = 0$ ($a, b \in \mathbb{Q}$) and $2x^3 + 5x^2 + 2x - 1 = 0$ may have a common root, then sum of all possible values of $(a - b)$ is
2. If θ is the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their point of intersection when $x > 0$, then $7 \tan \theta$ is equal to
3. Polynomial $P(x)$ contains terms of odd powers only. When $P(x)$ is divided by $x - 3$ and $x^2 - 9$ remainders are 6 and $g(x)$ respectively, then value of $g(2)$ is

Space for Rough Work

4. Consider curve $y = \sin(x - y)$, $-5\pi \leq x \leq 6\pi$ then sum of abscissa of points on the curve where tangent is perpendicular to line $2x + y = \pi$, is
5. If a cuboid, length of whose one edge is $\sqrt{2}$ units, is inscribed in a sphere of radius $\sqrt{2}$ units, then maximum possible volume of cuboid is : (in cubic unit)
6. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ where $a_i \in \mathbb{R}$ and $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2}{3x^3} \right)^{\frac{2}{x}} = e^4$, then $\sum_{i=0}^4 a_i$ is equal to

Space for Rough Work

7. If $n \in \mathbb{N}$ and $f(x) = \binom{n}{0}x + 1 \binom{n}{1}x + 2 \binom{n}{2}x + 3 \binom{n}{3}x + 4 \dots \binom{n}{n}x + n + 1$ and $f'(0) = \frac{3}{2}(7!)$,
then n is equal to
8. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x$, $\forall x \in \mathbb{R}$ then the
value of $|f''(1)|$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator